Lucas R. Kerbs

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Research Interests: Low-Dimensional Topology, Cannon-Thurston Maps, Geometric Group Theory.

Education

Washington University — Saint Louis, MO
PhD. in Mathematics — *Expected June 2027*
Advisor: Steven Frankel

Washington University — Saint Louis, MO
Masters of Arts in Mathematics — *Received June 2024*
Advisor: Steven Frankel

California Polytechnic University, San Luis Obispo — San Luis Obispo, CA
Masters of Science in Mathematics — *Received June 2022*
Advisor: Ryan Tully Doyle

- Thesis: Searching for Holes in the Matrix Universe, or the Beginnings of Algebraic Topology in Free Analysis.
- Honors: With Distinction

Biola University — La Mirada, CA
Bachelor of Science in Mathematics — *Received May 2020*

Conference/Workshop Attendance

Saint Louis Topology Conference — Saint Louis, MO
Student Organizer — *Summer 2024*

SLMath: Summer School — Berkeley, CA
Topological and Geometric Structures in Low Dimensions — *Summer 2025*

SLMath: Topological and Geometric Structures in Low Dimensions — Berkeley, CA
Program Associate — *Spring 2026*

Teaching

Full Responsibility Instructor

Summer 2024:	Quantitative Reasoning
Summer 2023:	Quantitative Reasoning
Winter 2022:	Pre-Calculus
Fall 2021:	Pre-Calculus
Spring 2021:	Business Calculus
Winter 2021:	Pre-Calculus
Fall 2020:	Pre-Calculus

Teaching Assistant

Fall 2025:	Differential Equations
Fall 2024:	Calculus III
Fall 2023:	Calculus I

Academic Recognition

Edward N. Van Duyne Scholarship 2021
Awarded to Mathematics Majors who excel academically.

Charles J. Hanks Excellence in Mathematics Award 2022
Awarded for outstanding accomplishment and talent in mathematics.

Invited Talks

March 2027: Spring Topology and Dynamics Conference, Session on Geometric Group Theory
Universal Cantor Sets and Cannon–Thurston Maps for Free-by-Cyclic Groups

November 2026: University of Virginia, Geometry Seminar
Pinched Jordan Curves and Cannon-Thurston Maps

October 2026: AMS Central Sectional, Special Session on Geometry, Group Actions, and Dynamics
Universal Cantor Sets and Cannon–Thurston Maps for Free-by-Cyclic Groups

November 2022: Washington University in St. Louis, Szego Seminar
Searching for a Free Fundamental Group